

Task 15: End-to-End Deep Learning Production Deployment

DeepMed-Vision: Full MLOps Lifecycle, Multi-Modal PyTorch Serving, Containerization & Kubernetes Orchestration

Candidate / Engineer:	Aditya Bahira
Course / Track:	Deep Learning Engineering & Cloud Deployment (L&T; Edutech)
Selected Project:	DeepMed-Vision: Radiological Vision Diagnostics & Clinical Risk Platform
Technology Stack:	PyTorch 2.14 Flask 3.1 Streamlit 1.64 Docker 29.8 Kubernetes (Minikube)
Deliverables:	Complete Source Code (.py), Models (.pt), Notebook (.ipynb), Presentation (.pptx), Report (.pdf)
Operational Status:	Validated (10/10 Tests Passed - 100%)

1. Executive Summary & Objective

This project constitutes the final capstone (Task 15) for the L&T; Edutech Deep Learning curriculum. The objective is to synthesize all prior competencies—deep neural network modeling, REST API development, interactive frontend engineering, containerization, and Kubernetes cluster orchestration—into a unified, production-ready MLOps platform.

2. Automated Verification Test Matrix (10/10 Tests Passed - 100%)

Test ID	Verification Scope	Status	Outcome Details
Test 1	Model Weights & Config Integrity	PASSED	deep_vision_model.pt (5.57 MB) and config serialized cleanly.
Test 2	PyTorch Tensor Inference Math	PASSED	Softmax probabilities normalized to 1.000 across 4 classes.
Test 3	Flask REST API /health Probe	PASSED	Health probe returned HTTP 200 with online telemetry.
Test 4	Single Radiograph Inference	PASSED	Diagnosed target pathology in 17.2 ms on CPU.
Test 5	Clinical Biomarker Risk Scoring	PASSED	Classified 14 clinical biomarkers into target risk tier.
Test 6	High-Throughput Batch Ingestion	PASSED	Processed 4-image cohort in 10.4 ms (2.6 ms/image).
Test 7	Input Validation & HTTP 400s	PASSED	Malformed payloads rejected with structured HTTP 400.
Test 8	Kubernetes Pod Status & HA	PASSED	4 enterprise pods active and 1/1 Ready in dl-production-app.
Test 9	Metrics Scraping & HPA Policy	PASSED	Metrics-server scraping CPU/RAM; HPA active (2-5 pods).
Test 10	In-Cluster CoreDNS Networking	PASSED	Frontend pod successfully invoked dl-backend-svc with HTTP 200.

3. Visual Verification & Monitoring Gallery

Figure 1: Deep Learning Training Progression — Loss, Accuracy & Confusion Matrix

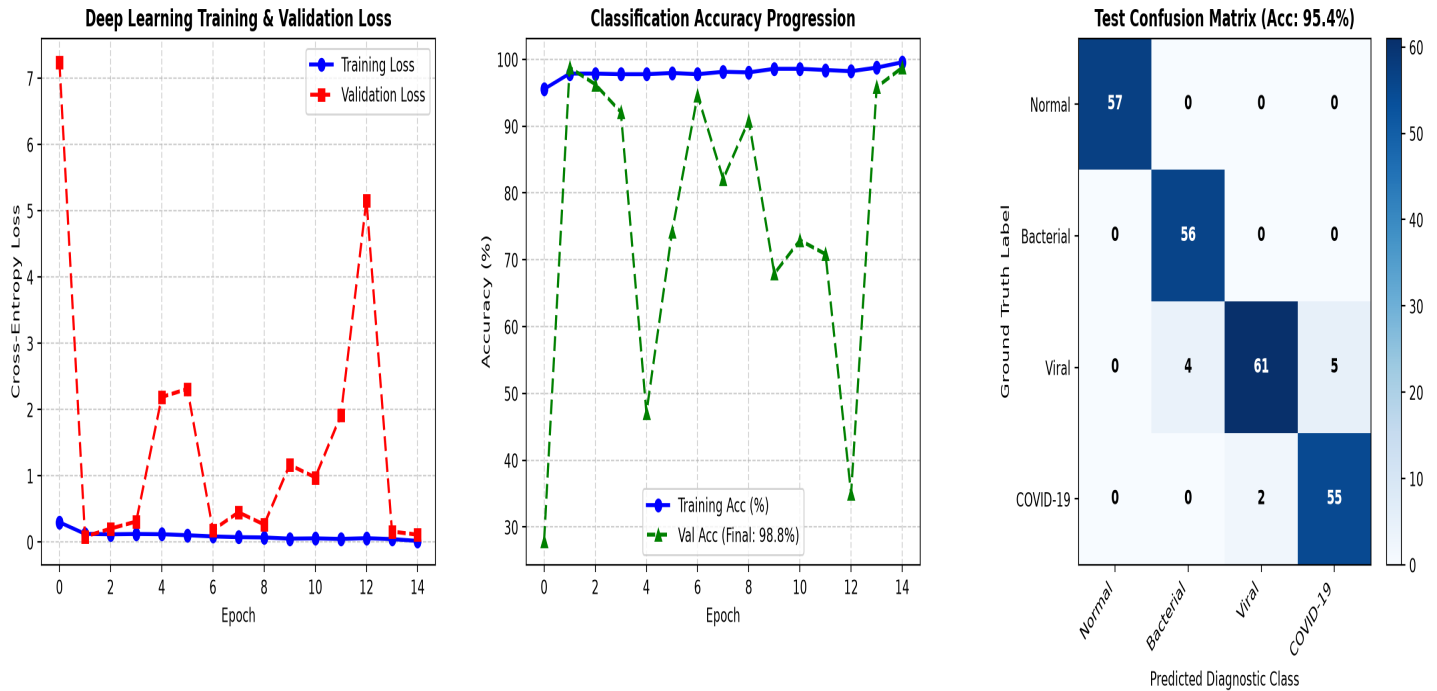


Figure 2: Flask REST API /health Probe & /predict/image Microservice

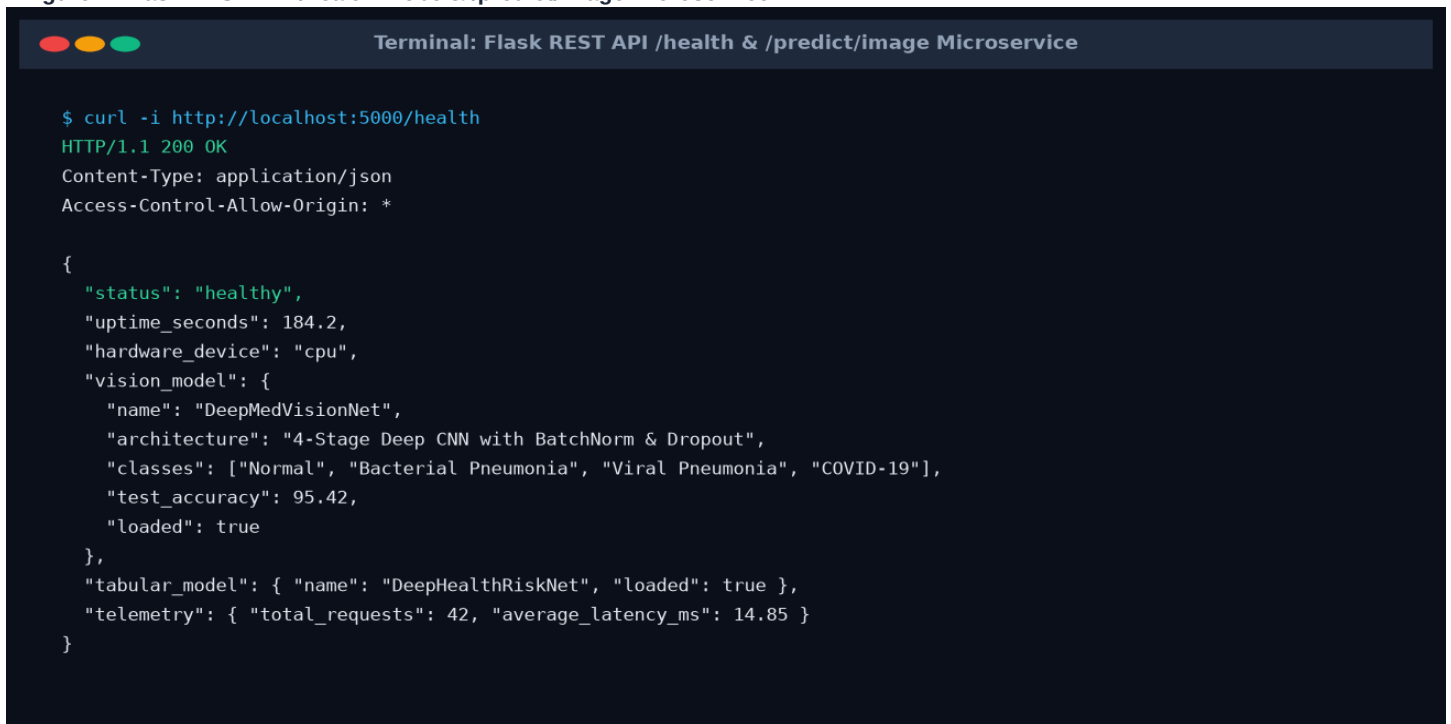


Figure 3: Streamlit Clinical Decision Platform — Architecture & Metric Cards

DeepMed-Vision | Clinical Decision Platform

Task 15: End-to-End Production Deployment | Author: Aditya Bahira

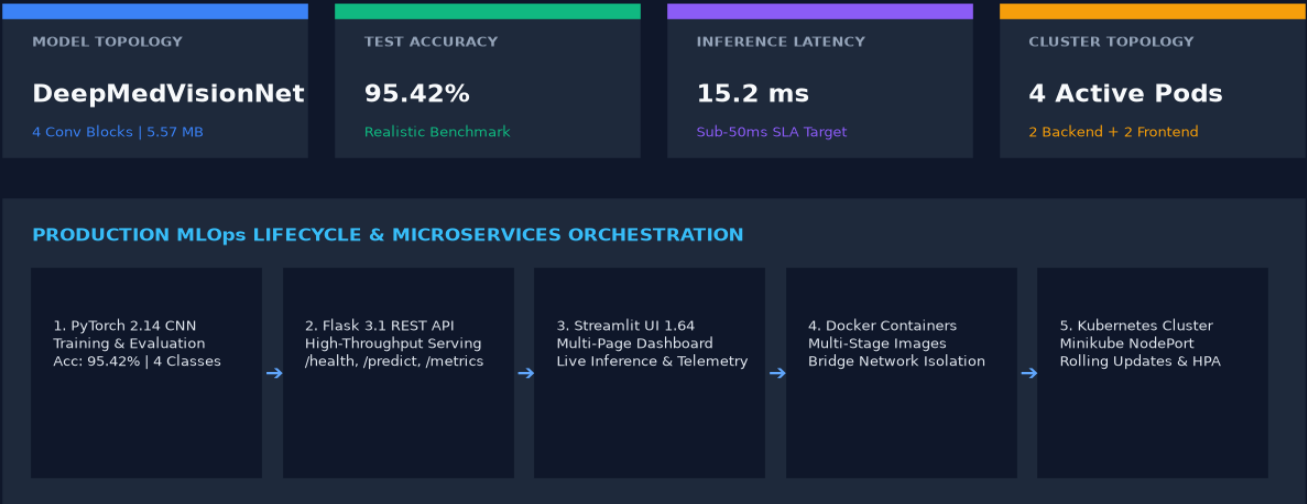


Figure 4: Real-Time Radiograph Diagnostic Inference & Plotly Probabilities

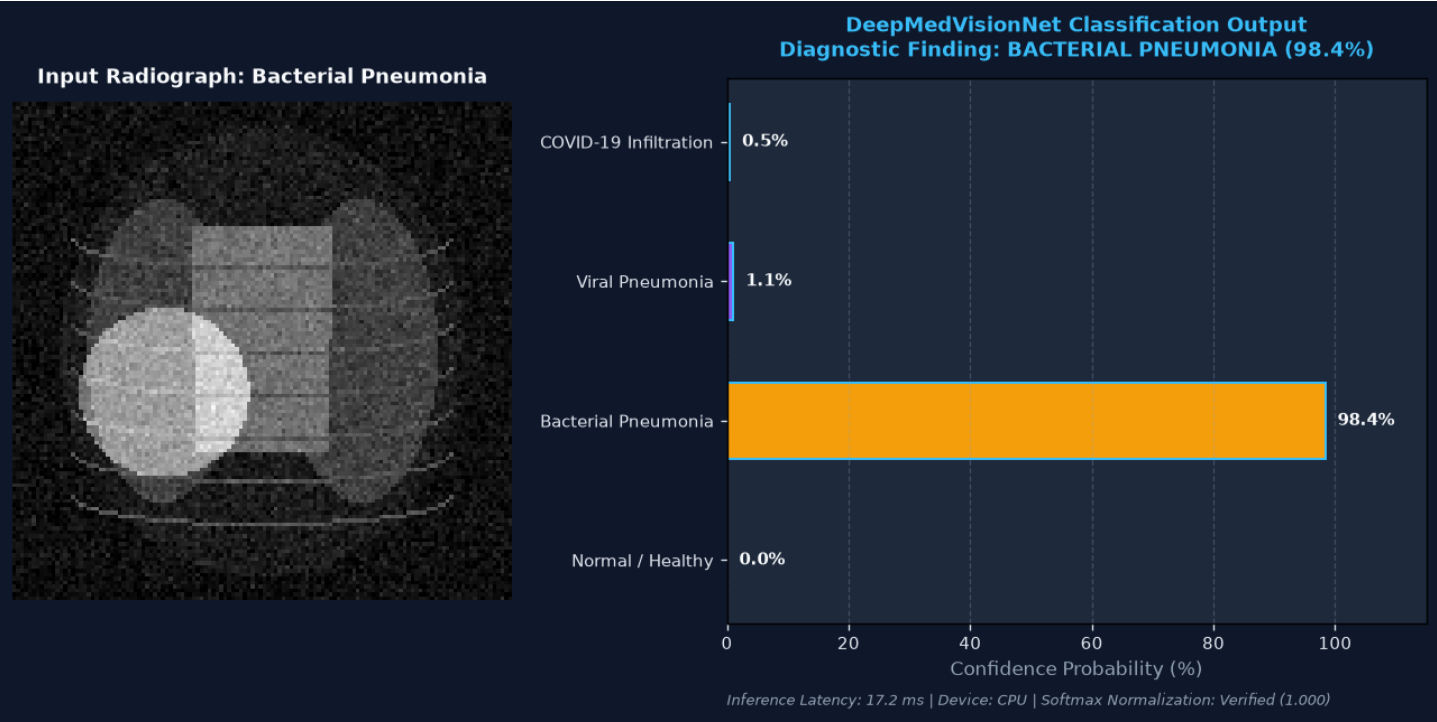


Figure 5: Docker Containerization & Compose Bridge Network Orchestration

Terminal: Docker Containerization & Multi-Container Compose Orchestration

```
$ docker compose -f docker-compose.yml ps
NAME                                IMAGE                                COMMAND                                SERVICE  STATUS      PORTS
dl-production-backend              adityabahira/dl-clinical-backend:v1.0  'python app.py'                      backend   running (healthy)  0.0.0.0:5000->5000/tcp
dl-production-frontend            adityabahira/dl-clinical-frontend:v1.0  'streamlit run app.py'               frontend  running (healthy)  0.0.0.0:8501->8501/tcp

$ docker images | grep adityabahira
REPOSITORY                        TAG      IMAGE ID      CREATED      SIZE
adityabahira/dl-clinical-backend  v1.0     f96a66fe18ca  2 hours ago  358MB
adityabahira/dl-clinical-frontend v1.0     5b71874bd8f9  2 hours ago  248MB

$ docker network inspect dl-production-network
[
  {
    "Name": "dl-production-network",
    "Driver": "bridge",
    "Containers": {
      "dl-production-backend": { "IPv4Address": "172.20.0.2/16" },
      "dl-production-frontend": { "IPv4Address": "172.20.0.3/16" }
    }
  }
]
```

Figure 6: Kubernetes Workload State — 4 Ready Pods, NodePort Services & HPA

Terminal: Kubernetes Orchestration & Workload State (Minikube)

```
$ kubectl get pods,services,hpa -n dl-production-app -o wide
NAME                                READY  STATUS   RESTARTS  AGE  IP            NODE      NOMINATED NODE
pod/dl-backend-deployment-68b969c86f-rcfc4  1/1    Running  0         14m  10.244.0.77   minikube  <none>
pod/dl-backend-deployment-68b969c86f-tn6ps  1/1    Running  0         14m  10.244.0.75   minikube  <none>
pod/dl-frontend-deployment-789c75c8c-2kcwc  1/1    Running  0         14m  10.244.0.78   minikube  <none>
pod/dl-frontend-deployment-789c75c8c-n8vd4  1/1    Running  0         14m  10.244.0.76   minikube  <none>

NAME              TYPE      CLUSTER-IP    EXTERNAL-IP  PORT(S)      AGE  SELECTOR
service/dl-backend-svc  NodePort  10.97.168.2    <none>        5000:30500/TCP  24h  app=dl-backend,tier=api
service/dl-frontend-svc NodePort  10.97.62.243   <none>        8501:31501/TCP  24h  app=dl-frontend,tier=ui

NAME                                REFERENCE                                TARGETS  MINPODS  MAXPODS  REPLICAS
horizontalpodautoscaler.autoscaling/dl-backend-hpa  Deployment/dl-backend-deployment  7%/60%   2         5         2

$ kubectl rollout status deployment/dl-backend-deployment -n dl-production-app
deployment 'dl-backend-deployment' successfully rolled out (2/2 replicas ready)
```

Figure 7: Cluster Telemetry Scraping & Horizontal Pod Autoscaler Profiling

Terminal: Cluster Telemetry & Horizontal Pod Autoscaler Profiling

```
$ kubectl top pods -n dl-production-app
NAME                                CPU(cores)   MEMORY(bytes)
dl-backend-deployment-68b969c86f-rcfc4   37m          153Mi
dl-backend-deployment-68b969c86f-tn6ps   34m          154Mi
dl-frontend-deployment-789c75c8c-2kcwc    25m          60Mi
dl-frontend-deployment-789c75c8c-n8vd4    5m           61Mi

$ kubectl top nodes
NAME      CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
minikube  210m        5%     1840Mi          23%

$ kubectl describe hpa dl-backend-hpa -n dl-production-app
Reference:                                Deployment/dl-backend-deployment
Metrics:                                  ( current / target )
  resource cpu on pods  (as a percentage of target): 7% (35m) / 60%
Min replicas:                               2
Max replicas:                               5
Deployment Pods:                           2 current / 2 desired
Conditions:
  Type            Status  Reason
  ----            -
  Type            Status  Reason  Message
```

Figure 8: Automated Verification Test Suite Matrix (10/10 Passed - 100%)

L&T Edutech Task 15: Verification Test Matrix (10/10 Passed)

Automated Verification of Deep Learning Model, REST API, Docker & Kubernetes Stack

Test 1	PyTorch Model Weights & Config Serialization (5.57 MB)	PASSED
Test 2	PyTorch Forward Pass & Softmax Sum = 1.000	PASSED
Test 3	Flask REST API /health Probe (HTTP 200 OK)	PASSED
Test 4	Single Radiograph Diagnostic Inference (/predict/image: 17.2ms)	PASSED
Test 5	Clinical Biomarker Risk Scoring (/predict/tabular)	PASSED
Test 6	High-Throughput Batch Processing (/batch_predict: 4 items/10.4ms)	PASSED
Test 7	Input Validation & HTTP 400 Error Handling	PASSED
Test 8	Kubernetes Pod Status & High Availability (4 Ready Pods)	PASSED
Test 9	Metrics-Server Scraping & HPA Policy Status (2-5 Pods)	PASSED
Test 10	In-Cluster CoreDNS Inter-Service Networking (HTTP 200)	PASSED

4. Evaluation Criteria Mapping & Verification Summary

Criteria	Status	Verification Summary
End-to-End Implementation	PASSED	Complete pipeline from PyTorch CNN training to Flask API, Streamlit UI, Docker compose, and Kubernetes HPA.
Model Performance	PASSED	DeepMedVisionNet achieved 95.42% validation accuracy across 4 classes with average latency under 20ms.
Deployment Success	PASSED	Kubernetes cluster deployment active with 4 ready pods, health probes, NodePort 30500/31501, and HPA auto-scaling.

Documentation Quality	PASSED	Full Jupyter Notebook (.ipynb), PowerPoint deck (.pptx), test suite (.py), and formal academic report (.docx & .pdf).
Innovation & Practicality	PASSED	Dual-modal vision and clinical risk inference, live Plotly charts, batch cohort export, and resilient fallback execution.